

Cloud Computing Lab 5: Persistent Volumes and Stateful Sets in Kubernetes

Due Date

Wednesday, October 29, 2025 11:59 PM

Objectives

In this lab, we will be returning to Minikube to deploy a Postgres database storing data in pods with Persistent Volumes (PVs) and managing pods with StatefulSets.

Overview

This assignment focuses on new characteristics of Kubernetes: persistent volumes and stateful sets. The PVs will be used for storage; stateful sets will manage your pods in place of the deployment.yaml file you created in Lab 4. Because of the cost charge associated with EC2s, we will be using Minikube for this assignment, so both the pods and the storage will be stored locally on your computer.

DIY

This assignment differs from lab 4 in that the storage of your server should connect to your persistent volume in your Kubernetes cluster, so you will store text strings in your PV. Setting up and configuring your PV will include creating a PV and a PVC (persistent volume claim) and creating/applying the matching yaml files; these yaml files should also configure your cluster to recognize Postgres and allow you to use Postgres commands on terminal to save data. Likewise, to manage your PV, you will need to apply a statefulset to your application. Instead of creating and applying a deployment.yaml file, you should do the same for a statefulset.yaml file.

To help you organize your work for this lab, here are the major checkpoints:

- **Checkpoint 1 - Containerization:**

- Create Dockerfile for your Postgres database
- Build and test Docker image locally
- Push to Docker Hub and verify Minikube can pull it

- **Checkpoint 2 - Persistent Storage Setup:**

- Create and apply PV YAML configuration for local storage
- Create and apply matching PVC YAML
- Configure yaml files so that cluster can recognize Postgres and is compatible with the mechanism
- Test storage functionality locally by running Postgres commands in terminal

- **Checkpoint 3 - StatefulSet Deployment:**

- Create StatefulSet YAML to manage your pods
- Configure proper storage mounts and pod specifications
- Deploy and verify application functionality
- Test persistent storage of favorite responses
- **Reference Resources:**
 - Persistent Volumes on Kubernetes
 - Understanding StatefulSets in Kubernetes
 - PostgreSQL on Kubernetes

Grading

Create a video demonstration (less than 2-minutes) showing the below. Submit your video demo along with all YAML configuration files.

- 15 pts - Minikube Configuration: Working local cluster with proper networking and Docker integration
- 15 pts - Postgres functionality: ability to create tables and store data when sending Postgres commands in terminal
- 25 pts - Persistent Volume Setup: Correctly configured and mounted PV/PVC with proper YAML files; when a pod is deleted and restored, the data stored in the pod should persist
- 25 pts - StatefulSet Implementation: Proper pod management and deployment using StatefulSet configuration
- 20 pts - Video Demonstration: Show working cluster, ability to store data using Postgres commands, PV/StatefulSet functionality

Submission

Submit this assignment to your Lab 5 repo on GitHub before the due date.